Acne Detection and Cross-Domain Classification — Report

1 Overview

This project tackles two connected problems on dermatology imagery: (1) detecting individual acne lesions in ACNE04 with bounding-box object detection, and (2) transferring an acne classifier trained on ACNE04 patches to DermNet, a clinical skin-condition dataset with a substantial domain gap. Part 1 measures localization quality on a single dataset; Part 2 measures how well learned acne features generalize across imaging conditions, framing, and label distributions.

2 Evaluation metrics

For detection: IoU $= |B_{\text{pred}} \cap B_{\text{gt}}|/|B_{\text{pred}} \cup B_{\text{gt}}|$, Precision $= TP/(TP+FP)$, Recall $= TP/(TP+FN)$; for DermNet: $F_1 = 2PR/(P+R)$. I report **mAP** (COCO-style), precision/recall, and mean matched IoU@0.5 for detection; **Accuracy**, **F1**, and **AUROC** for classification.

3 Part 1: Object Detection on ACNE04

3.1 Models

I trained two detectors spanning classic vs. modern: **Faster R-CNN** (Ren et al., 2015) — two-stage, ResNet-50 FPN, COCO-pretrained — and **YOLOv5s** (Ultralytics) — single-stage YOLO lineage. YOLOv5 was chosen over heavier detectors (e.g. DINO-DETR) for efficiency and strong community baselines on small objects. YOLOv5 uses native multi-class labels (papules, pustules, whitehead/blackhead); Faster R-CNN uses one "acne" class; box-level mAP/IoU remain comparable.

3.2 Data, training, evaluation

ACNE04 (Roboflow): YOLO + COCO exports for fair splits. Patched `data.yaml` with absolute paths. **YOLOv5:** 50 epochs, 640px, batch 16, `yolov5s.pt`. **Faster R-CNN:** 10 epochs, SGD lr 0.005, momentum 0.9, wd 5×10^{-4}, batch 4, two classes. Evaluated on COCO test with `torchmetrics.MeanAveragePrecision`; greedy matching (conf >0.25, IoU ≥ 0.5) for Precision@0.5, Recall@0.5, mean matched IoU@0.5. YOLO metrics from `results.csv`; matched IoU recomputed via `torch.hub` for parity.

3.3 Visualization

Figure 1 shows six test images per detector. YOLOv5 shows per-subtype labels; Faster R-CNN shows red boxes with confidences mostly 0.6–0.95 (higher recall in Table 1); YOLOv5 boxes appear slightly tighter (higher mean IoU).

4 Part 2: Cross-Domain Classification

4.1 Patches, baseline, DermNet setup

Positive/negative 64×64 patches from ACNE04 boxes (equal counts); box stats checked for scale. **Baseline:** ResNet-18 ImageNet (He et al., 2016), 10 epochs, Adam 10^{-4}, flips + color jitter, 80/20 split. DermNet binary: `Acne and Rosacea Photos` vs. all other folders (∼1:22 imbalance). `ImageFolder` order differs from ACNE04 (`negative/positive` ⇒ acne index 1); evaluation maps label spaces explicitly.

4.2 Domain adaptation

(1) `model_cls2`: InceptionResnetV1 VGGFace2 (`facenet-pytorch`), 160×160, stronger aug. (45° rot., jitter), class weights $[1.0, 5.0]$. (2) `model_cls3`: Per-channel histogram matching to random DermNet refs (from 20 unlabeled dev samples), then FaceNet retrain. (3) `model_cls4`: Pseudo-labels from `model_cls3` on 10 acne + 10 other train images; top-K=5 per class; 5 epochs fine-tune at 10^{-5}.

4.3 Evaluation

All checkpoints evaluated with `evaluate_on_dermnet`: resize per model, P(acne), F1-optimal threshold from PR curve; Accuracy, F1 (acne positive), AUROC.

5 Results

Table 1 summarizes Part 1. Similar mAP@0.5 (∼0.21–0.23); Faster R-CNN **recalls** more (0.45 vs. 0.28); YOLOv5 is more **precise** and slightly tighter boxes. Recall favors counting tasks; precision/speed favors real-time use.

Table 2 reports Part 2 on the full DermNet binary test set (accuracy, F1, AUROC, and F1-optimal threshold per checkpoint).

5.1 Grad-CAM

Grad-CAM on `model_cls2`–`model_cls4`: 10 DermNet test images each (5 acne / 5 non-acne), FaceNet `block8`, acne class. Figure 2: 2×5 panels; `True/Pred` with confidence; green/red = correct/incorrect; warm colors = high contribution to acne activation.

6 Analysis (reflection)

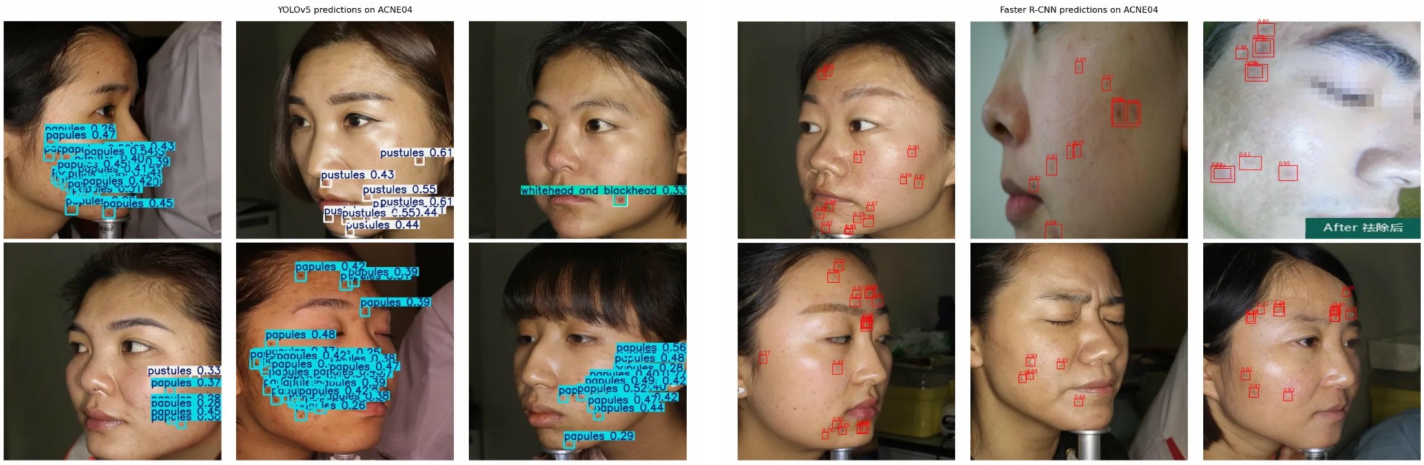
What helped. As Table 2 shows, histogram matching yielded best AUROC (0.6708), matching the photographic ACNE04→DermNet gap seen in the 20 unlabeled samples. FaceNet + aug. raised accuracy (0.7347) but *lowered* AUROC (0.5982) and pushed the F1 threshold to 0.96 — a conservative classifier that mostly wins on the 1:22 majority class. Pseudo-labeling on 20 images *hurt* vs. HM alone; too little signal.

Challenges / failure modes. Low F1 (∼0.2) everywhere: imbalance + patch→full-image gap. Heatmaps (Fig. 2) show diffuse or mis-localized attention; `cls4` can be sharp but wrong. Several panels attend near the **DermNet.com watermark** (shortcut learning). Thresholds vary widely (0.19–0.96) — poor calibration across runs.

Next steps. More pseudo-labels; DANN/CORAL-style alignment; balanced sampling / focal loss + PR-AUC; remove/crop watermarks.

References

1. Ren et al. (2015). Faster R-CNN. https://arxiv.org/abs/1506.01497
2. Redmon & Farhadi (2018). YOLOv3. https://arxiv.org/abs/1804.02767
3. Lin et al. (2014). MS COCO. https://arxiv.org/abs/1405.0312
4. He et al. (2016). ResNet. https://arxiv.org/abs/1512.03385
5. Selvaraju et al. (2017). Grad-CAM. https://arxiv.org/abs/1610.02391



a YOLOv5 **b** Faster R-CNN

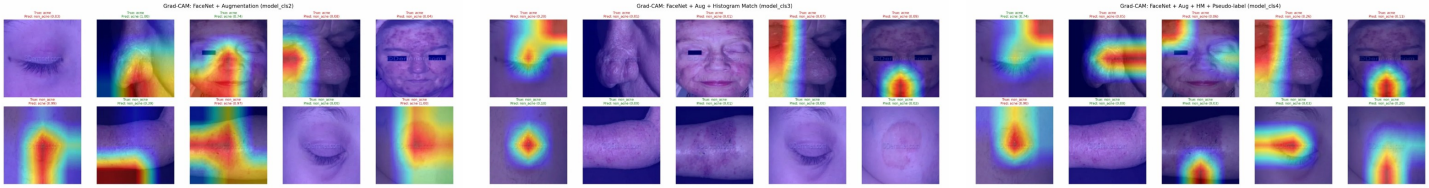
Figure 1: Qualitative detection on ACNE04 (six test images each).

Metric	YOLOv5	Faster R-CNN
mAP@0.5	0.2111	**0.2302**
mAP@0.5:0.95	**0.0520**	0.0507
Precision@0.5	**0.3068**	0.2851
Recall@0.5	0.2840	**0.4467**
Mean IoU@0.5 (matched)	**0.6323**	0.6218

Table 1: Detection on ACNE04 test (COCO split).

Model	Acc.	F1	AUROC	Thr.
Baseline (ResNet-18)	0.6661	0.2057	0.6471	0.3279
FaceNet + Aug.	0.7347	0.1841	0.5982	0.9592
Aug + Hist. match	0.7025	0.2076	**0.6708**	0.1944
Aug + HM + Pseudo	0.6397	0.2061	0.6477	0.2429

Table 2: DermNet binary test (acne vs. non-acne).



a `cls2` **b** `cls3` **c** `cls4`

Figure 2: Grad-CAM on DermNet (FaceNet `block8`, acne class).